

## Analysis First-Year Exam, August 2018, Part 1

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Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let  $(a_n)_{n=0}^{\infty}$  be a sequence of positive real numbers. Prove that

$$1/\limsup a_n = \liminf(1/a_n)$$

in the sense that if one side lies in  $(0, \infty)$  then so does the other and the two sides are equal.

2. Let  $(a_n)_{n=0}^{\infty}$  be a sequence of real numbers and  $A = \{a_n : n \geq 0\}$  its set of values. For  $p$  a real number, consider the two implications. One of these is true and the other false; give a proof for the true and a counterexample for the false.

(i) if  $p$  is a limit point of  $A$  then  $p$  is the limit of a subsequence of  $(a_n)_{n=0}^{\infty}$ ;

(ii) if  $p$  is the limit of a subsequence of  $(a_n)_{n=0}^{\infty}$  then  $p$  is a limit point of  $A$ .

3. Let  $X$  be a compact metric space and let  $f : X \rightarrow X$  be a continuous function. Prove that if  $f$  has no fixed points, then there exists  $a > 0$  such that  $d(x, f(x)) \geq a$  whenever  $x \in X$ . Show by example that the compact hypothesis cannot be dropped.

4. This problem has two parts.

(i) Prove that if  $c$  is any real number then the following subset of the Euclidean plane is connected

$$\{(x, y) : (x = \pm 1) \text{ or } (|x| < 1 \text{ and } y \geq c)\}.$$

(ii) Let  $C_0 \supseteq C_1 \supseteq \dots$  be a decreasing sequence of closed, connected subsets of a metric space. If the intersection  $\bigcap_{n \geq 0} C_n$  is nonempty, must it be connected? Proof or counterexample.

5. Let  $f : (0, \infty) \rightarrow \mathbb{R}$  be differentiable and assume that  $f'(t) \rightarrow 0$  as the real number  $t$  tends to infinity. Show that if  $f(n) \rightarrow \ell$  as the integer  $n$  tends to infinity, then  $f(t) \rightarrow \ell$  as the real number  $t$  tends to infinity.